

5.To perform basic data preprocessing such as handling missing values using the Titanic dataset.

```
In [1]: #Import Required Libraries
import pandas as pd
import numpy as np
```

```
In [2]: #Load the Dataset
df = pd.read_csv(r"C:\Users\Smit\Downloads\Titanic-Dataset.csv")
```

```
In [3]: df.head()
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500



```
In [4]: df.shape
```

Out[4]: (891, 12)

```
In [5]: df.columns
```

Out[5]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'], dtype='object')

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass         891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp          891 non-null   int64
7   Parch          891 non-null   int64
8   Ticket         891 non-null   object
9   Fare           891 non-null   float64
10  Cabin          204 non-null   object
11  Embarked       889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [7]: #Check Missing Values
df.isnull().sum()
```

```
Out[7]: PassengerId     0
Survived              0
Pclass               0
Name                 0
Sex                  0
Age                 177
SibSp                0
Parch                0
Ticket              0
Fare                 0
Cabin                687
Embarked             2
dtype: int64
```

```
In [8]: #Percentage of Missing Values
(df.isnull().sum()/len(df))*100
```

```
Out[8]: PassengerId     0.000000
Survived              0.000000
Pclass               0.000000
Name                 0.000000
Sex                  0.000000
Age                 19.865320
SibSp                0.000000
Parch                0.000000
Ticket              0.000000
Fare                 0.000000
Cabin                77.104377
Embarked             0.224467
dtype: float64
```

```
In [9]: #Fill Missing Values of Age  
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
In [10]: #Fill Missing Values of Embarked  
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

```
In [11]: #Remove Cabin Column  
df = df.drop("Cabin", axis=1)
```

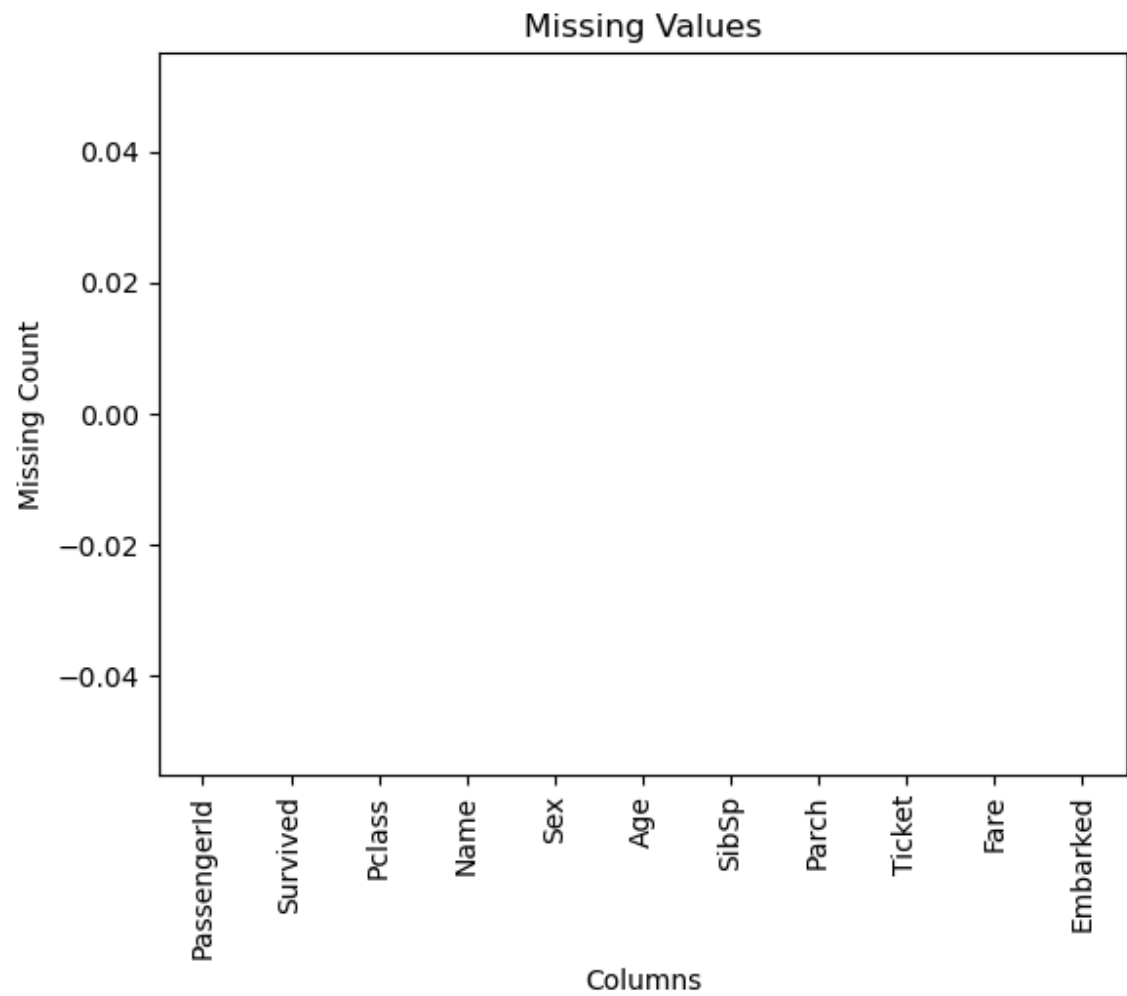
```
In [12]: #Verify Missing Values  
df.isnull().sum()
```

```
Out[12]: PassengerId    0  
Survived              0  
Pclass               0  
Name                 0  
Sex                  0  
Age                  0  
SibSp                0  
Parch                0  
Ticket              0  
Fare                 0  
Embarked             0  
dtype: int64
```

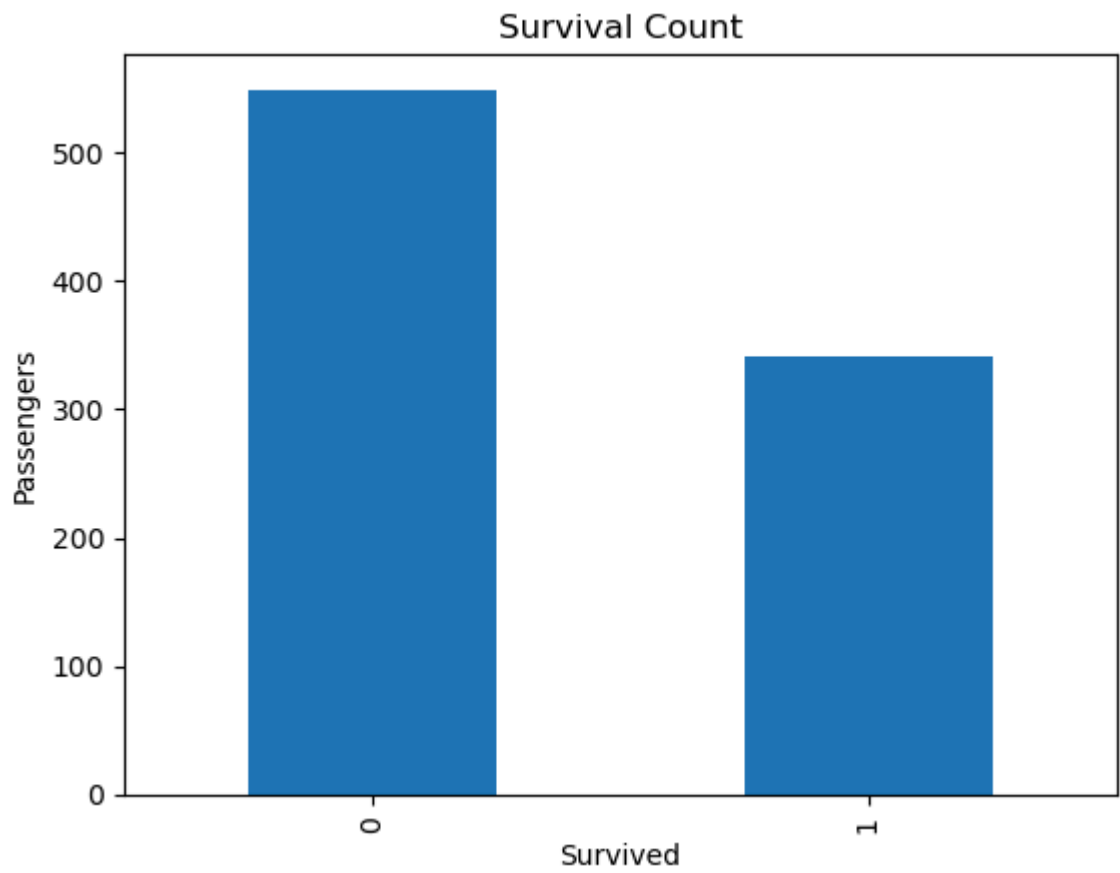
```
In [14]: #Save the Cleaned Dataset  
df.to_csv("Titanic_Cleaned.csv", index=False)
```

```
In [15]: #Missing Values Bar Chart
import matplotlib.pyplot as plt

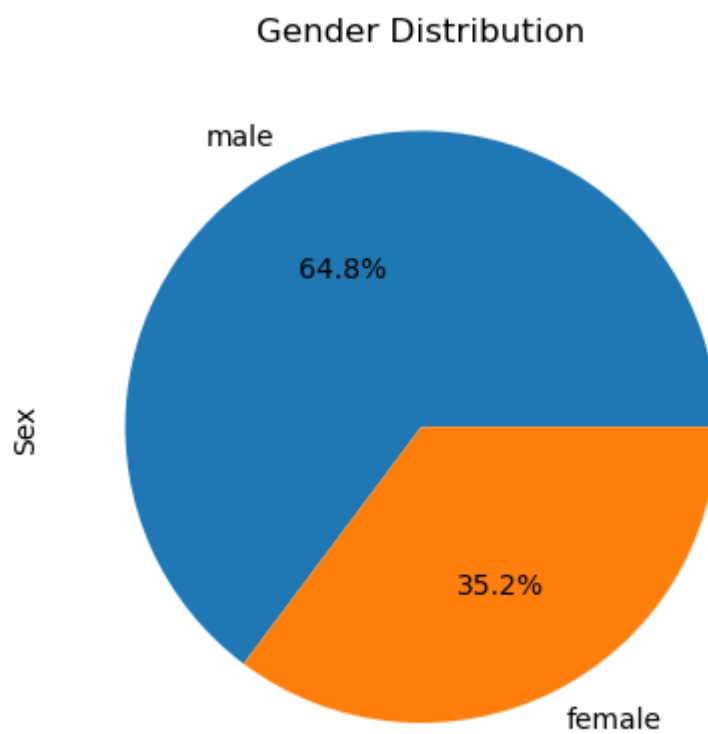
df.isnull().sum().plot(kind='bar')
plt.title("Missing Values")
plt.xlabel("Columns")
plt.ylabel("Missing Count")
plt.show()
```



```
In [16]: #Survival Count
df["Survived"].value_counts().plot(kind='bar')
plt.title("Survival Count")
plt.xlabel("Survived")
plt.ylabel("Passengers")
plt.show()
```



```
In [17]: #Gender Distribution
df["Sex"].value_counts().plot(kind='pie', autopct="%1.1f%%")
plt.title("Gender Distribution")
plt.show()
```



In []: